International Urology and Nephrology (2023) 55:17–28
https://doi.org/10.1007/s11255-022-03362-7

UROLOGY - ORIGINAL PAPER

Inhibitory effects of *Hydrocotyle ramiflora* on testosterone-induced benign prostatic hyperplasia in rats

Suyoung Park[1] · Youn-Hwan Hwang[2] · Eun-Bok Baek[1] · Eun-Ju Hong[1] · Young-Suk Won[3] · Hyo-Jung Kwun[1]

Received: 25 July 2022 / Accepted: 5 September 2022 / Published online: 15 September 2022

Abstract

Purpose Benign prostatic hyperplasia (BPH) is a urogenital disorder that affects approximately 85% of males who are over 50 years of age. *Hydrocotyle ramiflora* (HR), belonging to Apiaceae family, is used to treat urinary system diseases such as urine retention in traditional Chinese herbal medicine. In this study, we evaluated the effects of HR in the BPH animal model.
Methods We induced BPH in rats via subcutaneous (sc) injections of testosterone propionate (TP, 3 mg/kg). Rats were also administered HR (150 mg/kg), finasteride (10 mg/kg), or vehicle via oral gavage. After induction, prostate glands were collected, weighed, and processed for further analysis, including histopathological examination and immunohistochemistry. In addition, the mRNA expression of inflammatory cytokines in prostatic tissues was determined by quantitative real-time PCR (qRT-PCR). The protein expression of pro-apoptotic markers was examined using western blotting.
Results HR treatment significantly reduced the prostate weight, epithelial thickness, and proliferating cell nuclear antigen (PCNA) expression, with the levels of cleaved caspase-3 and Bcl-2-associated X (Bax) protein considerably increased compared to BPH group. HR also decreased inflammatory cell infiltration and pro-inflammatory cytokine levels compared with BPH group. Furthermore, the expression of phosphor-nuclear factor-κB (NF-κB), cyclooxygenase-2 (COX-2), and inducible nitric oxide synthase (iNOS) were reduced by HR treatment.
Conclusion These results indicate that HR suppresses the development of BPH associated with anti-proliferative, pro-apoptotic, and anti-inflammatory effects, suggesting it is a potential alternative therapeutic agent for BPH.

Keywords Apoptosis · Benign prostatic hyperplasia · *Hydrocotyle ramiflora* · Inflammation · Proliferation

Suyoung Park and Youn-Hwan Hwang have contributed equally to this work.

✉ Hyo-Jung Kwun
hyojung@cnu.ac.kr

1 Department of Veterinary Pathology, College of Veterinary Medicine, Chungnam National University, 99, Daehak-ro, Yuseong-gu, Daejeon 34134, South Korea

2 Herbal Medicine Research Division, Korea Institute of Oriental Medicine, 1672, Yuseong-daero, Yuseong-gu, Daejeon 34054, South Korea

3 Laboratory Animal Resource Center, Korea Research Institute of Bioscience and Biotechnology, 30, Yeongudanji-ro, Cheongwon-gu, Cheongju 28116, South Korea

Introduction

Benign prostatic hyperplasia (BPH) is a non-cancerous urologic problem, in which the growth of stromal and epithelial cells induces urethral obstruction [1]. The proliferation of prostate cells creates the enhanced prostatic tissue volume and increased stromal smooth muscle tone, thereby increasing urethral resistance, blocking the urethra and obstructing the urinary bladder [2, 3]. Due to these pathological changes, patients with BPH show symptoms such as lower urinary tract syndrome (LUTS), urinary tract infection, and hematuria [4].

The etiology of BPH is still unknown. The risk factors of BPH are classified as unalterable (age, genetics, and geography) and alterable (sex hormones, metabolic syndrome, inflammation, and diet) [3]. Particularly, sex steroid hormones causing changes in prostate development have been recognized to play important roles in the pathogenesis of BPH [5]. Through 5-α reductase enzyme, the prostate converts circulating testosterone into dihydrotestosterone (DHT). Subsequently, DHT interacts with androgen receptors and triggers protein synthesis and prostate cells proliferation [6]. The testosterone levels in the blood drop in healthy men as they age; on the contrary, DHT secretion is considerably enhanced in patients with BPH [7, 8]. Elevated serum DHT concentration and DHT to testosterone ratios have been related to an enlarged prostate volume and a high BPH prevalence [9].

Recent research has revealed that inflammation plays an important role in the etiology of BPH. Indeed, the prostate is an immunocompetent gland with a limited population of inflammatory cells. However, certain triggers, such as hormonal shifts, infections (bacterial and viral), immunological reactions, urine reflux within the prostate, and metabolic syndrome, can induce an inflammation of the prostate in elderly men [10]. With a combination of one or more of these triggers, the number of inflammatory cells infiltrating the prostate is enhanced, leading to epithelial and stromal tissue destruction. Such a prostatic injury can activate cytokine release and induce the chronic wound healing process that increases growth factor secretion, which results in prostatic enlargement, leading to local hypoxia in response to higher oxygen demand [11]. Consequently, reactive oxygen species are released under hypoxic condition [12], promoting angiogenesis and the secretion of more growth factors. This mechanism establishes a vicious cycle that contributes to a continuous increase in prostate volume [10, 13].

Species of *Hydrocotyle* (Apiaceae family) are used in Taiwan folk medicine for treating common cold, menstrual pain, and jaundice [14]. *Hydrocotyle ramiflora* (HR), a perennial herb found in Korea, is known for its anti-inflammatory property. In traditional Chinese herbal medicine, it is used to treat bruising, jaundice, and retention of urine [15]. According to Kwon et al., some of the compounds that have been extracted from HR include alpha-spinasterol, capsidiol 3-acetate, monogalactosylmonoacylglycerols, and digalactosyldiacylglycerol. Among these, alpha-spinasterol promotes apoptosis of U937 cells [16], ameliorates the progression of diabetic nephropathy in streptozotocin-induced diabetic mice [17], and exerts an anti-inflammatory effect in mice with lipopolysaccharide (LPS)-induced inflammation [18]. Capsidiol is an anti-fungal phytoalexin [19] shown to have immunomodulatory and anti-neuroinflammatory effects in a mouse model of experimental autoimmune encephalomyelitis [20]. However, no scientific investigation has been conducted to evaluate the effect of HR on BPH. Therefore, in the current study, we aimed to examine the potential of HR as a treatment for BPH using an animal model of testosterone-induced BPH.

Materials and methods

Plant extracts

The extracts of HR were obtained from The Korea Plant Extract Bank at the Korea Research Institute of Bioscience and Biotechnology (Daejeon, Korea).

Ultrahigh-performance liquid chromatography–tandem mass spectrometry (UHPLC–MS/MS) analysis

UHPLC–MS/MS analysis was performed to characterize the phytochemicals in HR. HR (20 mg/mL) and reference standards (10 µg/mL) were dissolved in methanol and filtered using a syringe filter (0.2-µm pore size). The samples were analyzed using a Dionex UltiMate 3000 system equipped with a Thermo Q-Exactive mass spectrometer (Thermo Fisher Scientific, Bremen, Germany), according to a previously reported method [21]. Briefly, chromatographic separation was achieved using an Acquity BEH C18 1.7 µm column (100 mm × 2.1 mm i.d) (Waters Corp., Milford, MA, USA) under gradient elution with a mobile phase consisting of 0.1% (v/v) formic acid in water (A) and acetonitrile (B). The injection volume and flow rate were 3 µL and 0.25 mL/min, respectively. MS analysis was conducted using an electrospray ionization source with full MS-ddMS2 mode in both positive and negative ionization modes. The MS/MS parameters were set as follows: ion spray voltage, 3.8 kV; capillary temperature, 320 °C; scan range, 100–1500 m/z; resolution, 70,000. Data acquisition and processing were performed using Xcalibur v.3.2 and TraceFinder software (version 4.0; Thermo Fisher Scientific).

Experimental animals

Male Sprague–Dawley (SD) rats were obtained from Orient Bio (Seongnam, South Korea). The rats weighed 150–220 g and were 7-week-old. The rats were maintained under standard laboratory conditions (relative humidity: 50% ± 5%, temperature: 22 ± 2 °C, 12-h light/dark cycles) and were fed a standard rodent diet and water ad libitum. All protocols for the studies were approved by the Animal Experimental Ethics Committee of Chungnam National University (Daejeon, South Korea; Approval No. CNU-00711).

Animal study

The rats were allowed to acclimatize to the standard laboratory environment for one week before the start of the experiment. The number of total rats was twenty-eight and they were randomly divided into four groups ($n=7$). The negative control (NC) group rats were orally administered phosphate buffered saline (PBS) and subcutaneously (s.c.) injected with corn oil. PBS and corn oil were vehicle for the extract and testosterone, respectively. The BPH group rats were administered PBS by oral gavage and s.c. injected with 3 mg/kg body weight (BW) testosterone propionate (TP; Tokyo Chemicals Ins. Co, Tokyo, Japan) to induce BPH. The BPH + F group, a positive control group, were also induced with BPH by TP (s.c.) and orally administered 10 mg/kg BW finasteride, which is the medication for treating BPH. The BPH + HR group rats were s.c. administered TP and orally administered HR extract (150 mg/kg). The experimental period was four weeks, with the weight of each rat being measured weekly to modify treatments as per current BW. After the experiment, the rats were fasted for 16 h and euthanized using CO_2. Before being euthanized, blood samples were collected from the caudal vena cava and centrifuged to isolate serum, which was stored at − 80 ℃ for further analysis. The prostate glands of the rats were carefully trimmed from the connective tissue surrounding the organ tissues and weighed. The prostate samples were divided into three portions; one portion was fixed in 10% neutral buffered formalin (NBF) for staining and immunohistochemistry, and the others were snap-freezed by liquid nitrogen and stored at − 80 °C for western blotting and qRT-PCR analysis.

Enzyme-linked immunosorbent assay (ELISA)

Serum or prostatic testosterone and DHT levels were measured using commercial ELISA kits (ALPCO Diagnostics, NH, USA). All ELISA measurements were performed according to the manufacturer's instructions.

Hematoxylin and eosin staining

After necropsy, the prostate samples were fixed in 10% NBF, processed in a tissue processor, and embedded in a paraffin block. The paraffin block was cut into 4-µm sections and stained with hematoxylin and eosin (H&E) for histological observation. The ventral lobes of the prostates were observed for epithelial thickness under a light microscope (Nikon ECLIPSE Ni-U, Tokyo, Japan). Five areas (×200) per rat sample were chosen randomly and measured using NIS-element software (BR5.11, Nikon, Tokyo, Japan).

Immunohistochemistry

Immunohistochemistry analysis was performed to confirm the presence of proliferative cell nuclear antigen (PCNA; 1:2000, monoclonal, #ab29, Abcam, Cambridge, UK). The cut sections were deparaffinized, rehydrated, and treated with primary antibody against PCNA overnight in 4 ℃, followed by treatment with the Vectastain Elite ABC Mouse IgG kit (1:200, #PK-6102, Vector Laboratories, CA, USA) for 1 h according to the manufacturer's instructions. After the antibody phase, diaminobenzidine (DAB; #SK-4100, Vector Laboratories) was used to observe the immune complex, which was subsequently stained with hematoxylin. Five randomly selected regions (×200) were collected from samples of each rat, and PCNA-positive cells per 100 cells were counted.

Quantitative real-time PCR (qRT-PCR)

Using TRIzol reagent, RNA was extracted from the same amount (1.4 µg) of frozen prostate tissues and quantified. The total purified RNA was reverse transcribed using an RT kit (Toyobo). After reverse transcription, SYBR Green Master Mix (Thermo Scientific, MA, USA) was used for qRT-PCR. The data were interpreted using Applied Biosystems 7500 Real-Time PCR system software (Applied Biosystems, CA, USA). The specific primers used were as follows: 5-α reductase type 2, F: 5′-ATTTGTGTGGCAGAGAGAGG-3′, R: 5′-TTGATTGACTGCCTGGATGG-3′; tumor necrosis factor-alpha (*TNF-α*), F: 5′-GTCTGTGCCTCAGCCTCT TC-3′, R: 5′-CCCATTTGGGAACTTCGCCT-3′; interleukin (*IL*)-*6*, F: 5′-TAGTCCTTCCTACCCCAACT-3′, R: 5′-TTG GTCCTTAGCCACTCCTT-3′; *IL-8*, F′: 5′-CATTAATAT TTAACGATGTGGATGCGTTTCA-3′, R: 5′-GCCTAC CATCTTTAAACTGCACAAT-3′. PCR reactions included 10 min at 95 °C, followed by 40 cycles at 95 °C for 15 s (denaturing) and 60 °C for 1 min (annealing/extension). The relative expression of target genes was obtained using the $2^{-\Delta\Delta CT}$ formula using Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) as a housekeeping gene.

Western blotting

Prostate tissues were homogenized with RIPA lysis buffer (#9806, Cell signaling, MA, USA) and protein was extracted. Proteins were loaded in identical amounts (30 µg) on an SDS-PAGE gel and transferred onto nitrocellulose membranes. After blocking, the membranes were incubated with the primary antibodies overnight at 4 °C. The primary antibodies used were anti-caspase 3 (1:1000, polyclonal, #9662, Cell signaling technology), anti-Bax (1:200, monoclonal, #SC-7480, Santa Cruz biotechnology, TX, USA), anti-phospho-NF-κB (1:1000, monoclonal, #3033, Cell signaling), anti-total NF-κB (1:1000, polyclonal, #ab16502, Abcam), anti-COX-2 (1:1000, monoclonal, #12282, Cell signaling), anti-iNOS (1:1000, monoclonal, #ab136918, Abcam), and anti-β-actin (1:3000, monoclonal. #A5441, MilliporeSigma, MA, USA). The membranes were then probed with secondary antibodies for 2 h at 22 °C, and proteins with antibodies were detected using a chemiluminescence detection kit and quantified using CS analyzer 4 analysis software (Atto, Tokyo, Japan).

Statistical analysis

All results are expressed as mean ± standard deviation (SD). Multiple comparisons were performed using the one-way ANOVA, followed by Dunnett's post hoc test ($P < 0.05$). Statistical analyses were performed using GraphPad Prism 6 (GraphPad Software, Inc., CA, USA).

Results

Identification of HR phytochemicals

UHPLC–MS/MS analysis was performed to characterize the phytochemicals in HR. In comparison with each reference standard, seven phytochemicals contained in HR were identified in both positive and negative ion modes. The retention time, precursor ion, MS/MS fragments, and peak of each phytochemical are summarized and illustrated in Table 1 and Fig. 1A, respectively. Flavonoids (i.e., quercetin-3-O-sophoroside, isoquercitrin, astragalin, and quercetin) were suitably ionized in the positive ion mode, whereas the other phenolics (chlorogenic acid and rosmarinic acid) were detected in the negative ion mode. The mass spectral patterns of HR phytochemicals were consistent with those of a previous study [22]. In the ion chromatograms of several phytochemicals (e.g., chlorogenic acid, isoquercitrin, and quercetin), multiple peaks with the same m/z values and different retention times were observed suggesting the presence of isomers and derivatives (Fig. 1B). Further studies are needed to confirm the structures of the identified HR phytochemicals.

Effect of HR on prostate weight in a BPH rat model

Relative prostate weight was calculated as the ratio of prostate weight to body weight. As shown in Table 2, the relative prostate weights were higher in the BPH group (0.381% ± 0.05%) than in NC group (0.191% ± 0.012%). However, treatment with finasteride (0.248% ± 0.039%) or HR (0.303% ± 0.033%) significantly reduced the relative prostate weight compared with that in the BPH group. The inhibition percent of prostate growth in the finasteride (BPH + F) and HR (BPH + HR) groups was 69.93% and 38.96%, respectively.

Effect of HR on the levels of testosterone, DHT, and 5-α reductase in a BPH rat model

As shown in Fig. 2A–C, the concentrations of serum or prostatic testosterone and DHT in the BPH group were higher than those in the NC group. Meanwhile, finasteride- and HR-treated rats showed decreased levels of testosterone and DHT. In addition, the mRNA expression of 5-α reductase was reduced in the finasteride- and HR-treated groups compared with that in the BPH group (Fig. 2D).

Effect of HR on prostate epithelial thickness in a BPH rat model

According to histological examination with hematoxylin and eosin (H&E) staining, the prostates of rats in the NC group exhibited normal tissue structure, with a single layer of epithelial cells (Fig. 3A). On the contrary, the prostates of rats with BPH showed glandular hyperplasia, which consisted of numerous layers of epithelial cells as well as a decreased glandular luminal area (Fig. 3A). The gland epithelium presented increased in-folding and was more elongated than the normal epithelium. The finasteride- and HR-treated rats exhibited decreased epithelial hyperplasia compared with rats in the BPH group (Fig. 3A). Furthermore, the epithelial thickness of the prostate tissue was considerably lower in the finasteride- and HR-treated groups than in the BPH group (Fig. 3B).

Table 1 Phytochemicals identified in HR by UPLC–MS/MS

No.	Phytochemical	Formula	R_t (min)	Adduct	Calculated (m/z)	Estimated (m/z)	Error (ppm)	MS/MS fragments (m/z)
1	Chlorogenic acid	$C_{16}H_{18}O_9$	5.18	$[M-H]^-$	353.0878	353.0870	−2.365	191
2	Quercetin-3-*O*-sophoroside	$C_{27}H_{30}O_{17}$	5.92	$[M+H]^+$	627.1556	627.1556	−0.002	303
3	Isoquercitrin	$C_{21}H_20O_{12}$	6.77	$[M+H]^+$	465.1028	465.1030	0.465	303
4	Astragalin	$C_{21}H_20O_{11}$	7.3	$[M+H]^+$	449.1078	449.1078	−0.116	287
5	Rosmarinic acid	$C_{18}H_{16}O_8$	8.09	$[M-H]^-$	359.0772	359.0765	−1.955	197, 179, 161
6	Quercetin	$C_{15}H_{10}O_7$	9.6	$[M+H]^+$	303.0499	303.0500	0.090	–

R_t retention time

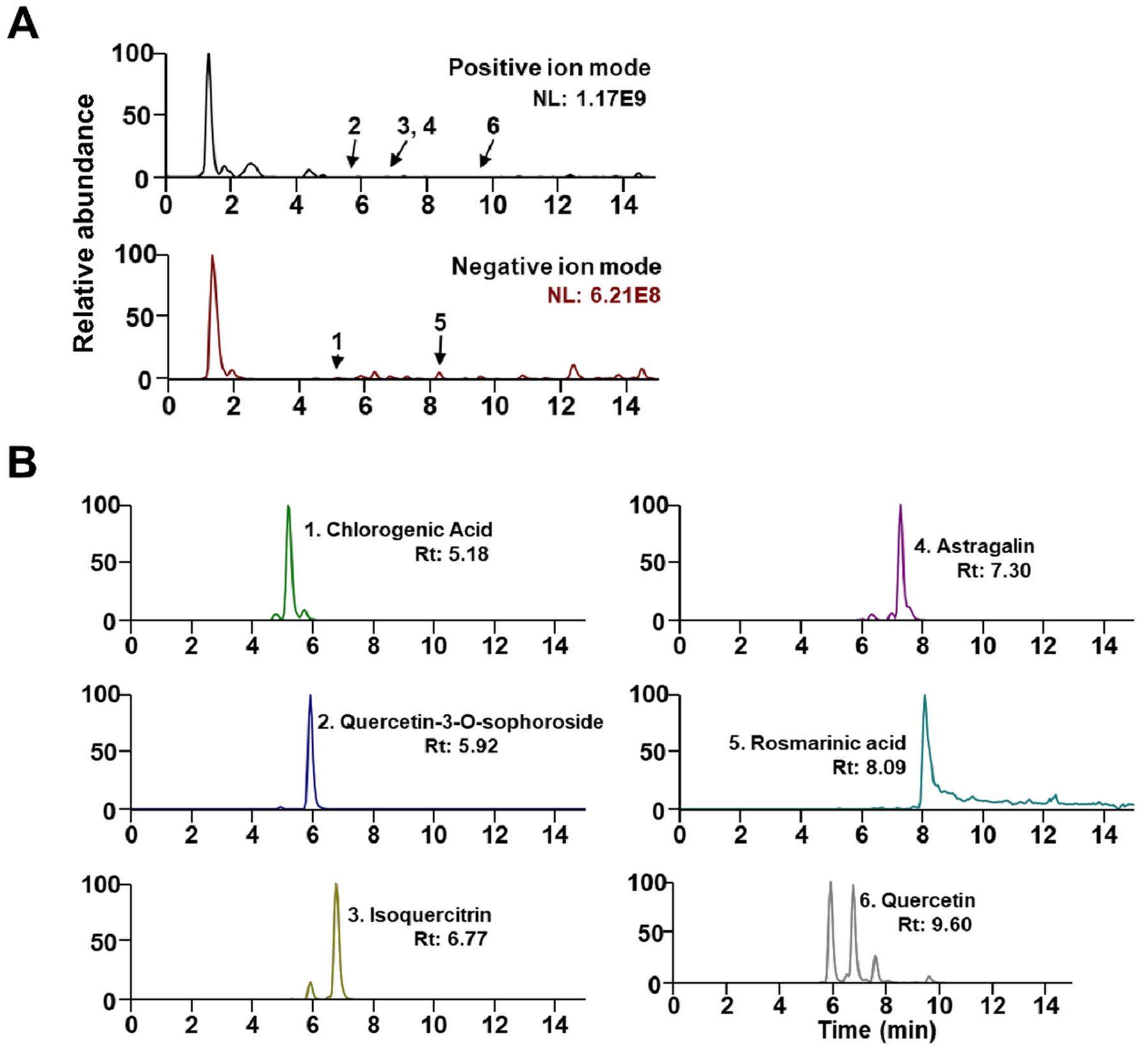

Fig. 1 UHPLC–MS/MS analysis of HR. **A** Base peak ion chromatograms in positive and negative ion modes. **B** Extracted ion chromatograms of the identified HR phytochemicals. *Rt* retention time

Table 2 Effect of HR on the prostate weight

Group	Treatment	Body weight (g)	Relative prostate weight (%)	% growth inhibition
NC	Corn oil + PBS	437.67 ± 19.33	0.191 ± 0.012	
BPH	TP + PBS	403.5 ± 14.5	0.381 ± 0.05##	
BPH + F	TP + Finasteride	425.33 ± 24.6	0.248 ± 0.039**	69.93
BPH + HR	TP + HR	420.8 ± 22.2	0.303 ± 0.033**	38.96

Relative prostate weight was calculated as prostate weight to body weight ratio. The percent growth inhibition was calculated as follows: % growth inhibition = 100 − [(treated group − negative control)/(positive control − negative control) × 100]. *NC* corn oil-administered (s.c.) and PBS-treated rats; BPH, TP (3 mg/kg)- and PBS-treated rats; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg)-treated rats; BPH + HR, TP (3 mg/kg) and HR (150 mg/kg)-treated rats

##$P < 0.01$ compared with the NC group

**$P < 0.01$ compared with the BPH group

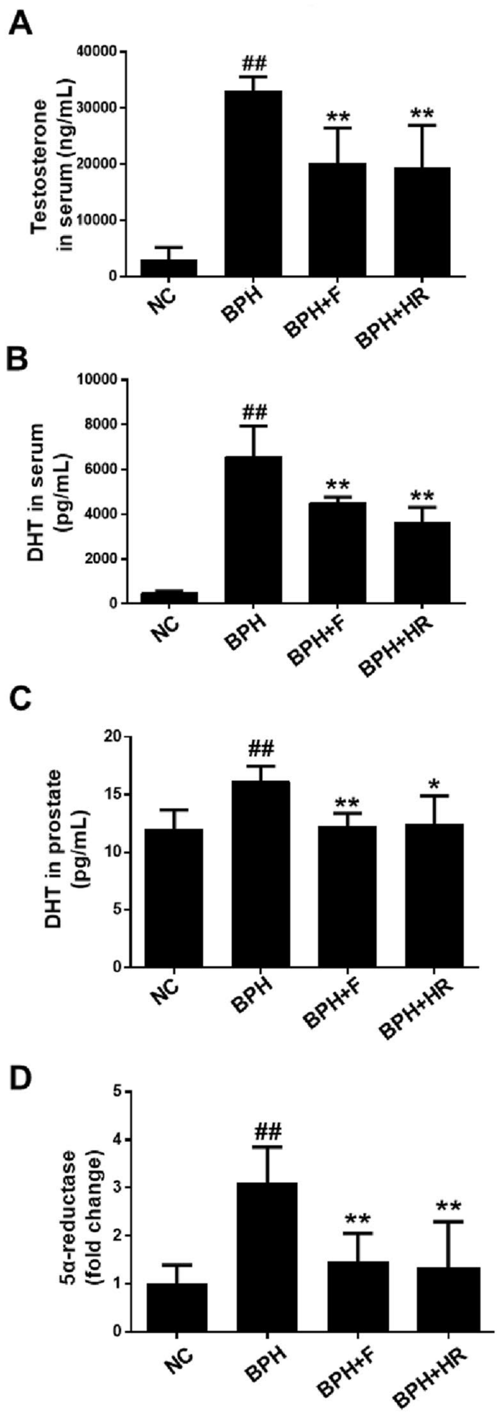

Fig. 2 Effects of HR on serum or prostatic levels of testosterone, DHT and 5α-reductase type 2 in rats with BPH. **A** Serum testosterone, determined using enzyme-linked immunosorbent assay (ELISA). **B** Serum DHT, determined using ELISA. **C** Prostatic DHT, determined using ELISA **D** Prostatic 5α-reductase type 2, determined using qRT-PCR. *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ compared with the NC group; $^{*}P < 0.05$ and $^{**}P < 0.01$ compared with the BPH group

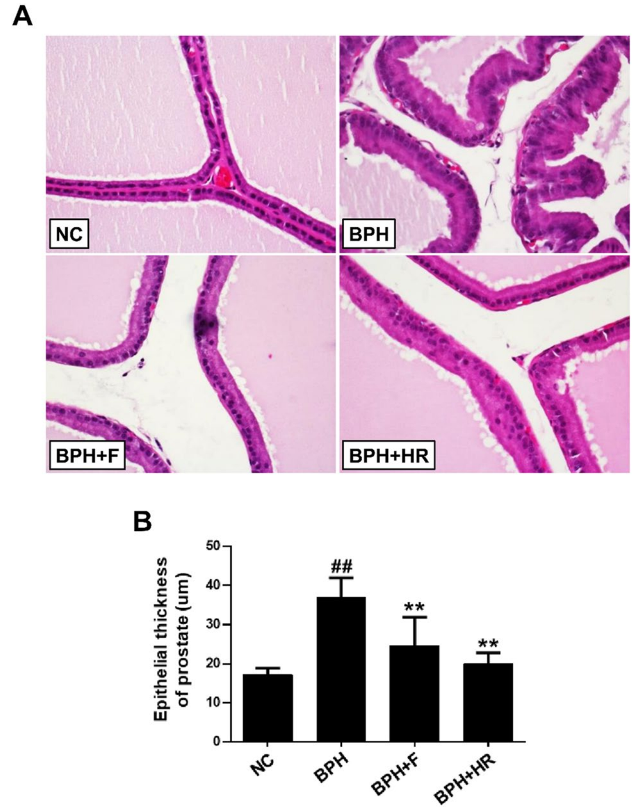

Fig. 3 Effect of HR on histological changes in prostate glands in rats with BPH. **A** Prostate tissue stained with H&E (final magnification, × 400). **B** Epithelial thickness of prostate cells (mm). *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ compared with the NC group; $^{*}P < 0.05$ and $^{**}P < 0.01$ compared with the BPH group

Effect of HR on prostate cell proliferation

Immunohistochemistry showed that proliferating cell nuclear antigen (PCNA)-positive nuclei were higher in the BPH group than in the NC group. However, finasteride and HR-treated group showed decreased number of PCNA-positive cells compared with the BPH group (Fig. 4).

Effect of HR on prostate cell apoptosis

Western blotting revealed that the expression of cleaved caspase-3 in the BPH group was lower than that in the NC group (Fig. 5A). Similarly, treatment with finasteride and HR significantly increased the levels of cleaved caspase-3 compared with the BPH group (Fig. 5A). Moreover, Bcl-2-associated X (Bax) expression was greater in the finasteride- and HR-treated groups than in the BPH group (Fig. 5B).

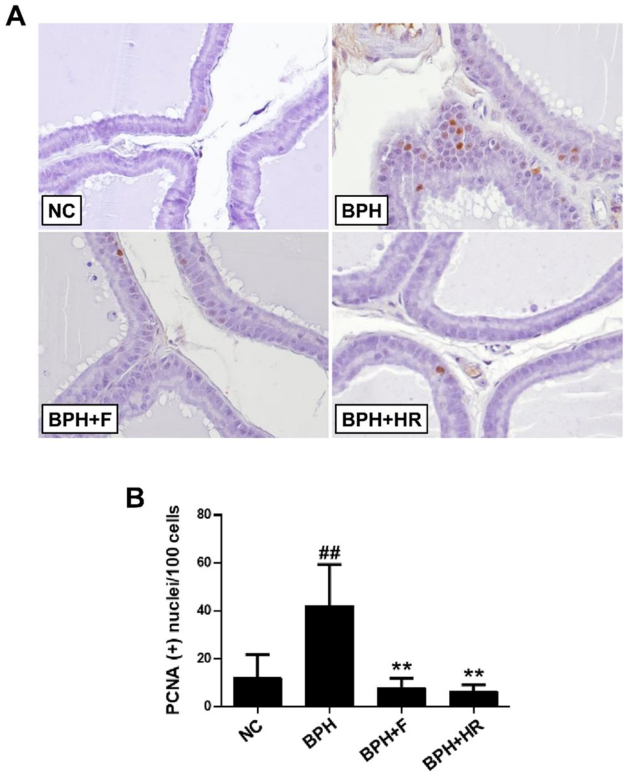

Fig. 4 Effect of HR on prostate epithelial cell proliferation in rats with BPH. **A** Immunohistochemical analysis of PCNA in prostate epithelial cells (final magnification, ×400). **B** PCNA-positive cells per 100 cells. *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ compared with the NC group; $^{*}P < 0.05$ and $^{**}P < 0.01$ compared with the BPH group

Effect of HR on inflammatory response in a BPH rat model

Following H&E staining, the BPH group showed infiltration of inflammatory cells, such as lymphocytes and mast cells, in the prostatic stroma, whereas the prostates of rats in the NC group showed no inflammatory cell infiltration (Fig. 6A). In addition, a few inflammatory cells were observed in the finasteride- and HR-treated rats (Fig. 6A). Similar to these results, the expression of tumor necrosis factor-alpha (TNF-α), interleukin (IL)-6, and IL-8 in the prostates was higher in the BPH group than in the NC group, but was significantly attenuated by finasteride and HR treatments (Fig. 6B).

Effect of HR on the expression levels of NF-κB, iNOS, and COX-2 in a BPH rat model

Phosphorylation of nuclear factor-κB (NF-κB) was increased in the BPH group. Conversely, the HR-treated rats showed lower levels of NF-κB activation than rats in the BPH group (Fig. 7A). In addition, increased levels of inducible nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2) following the induction of BPH were suppressed after finasteride and HR treatments (Fig. 7B and C).

Discussion

In the present study, we examined the therapeutic benefits of HR in a testosterone-induced BPH rat model. Compared with the BPH group, treatment of rats in HR group reduced the relative prostate weight and the epithelial thickness. Based on these findings, we deduced that HR hinders the development of testosterone-induced BPH in vivo.

The pathogenesis of BPH can be influenced by various factors [1, 3]. The DHT level increases by 3.5% annually in men aged 40–70 years, according to a longitudinal study, which coincides with the higher occurrence of BPH with increasing age [8, 23]. DHT binding in androgen receptors increases the number of cells in the prostate [24, 25], implying that DHT is a pivotal element in prostatic epithelial and stromal cell hyperplasia. It has been reported that 5-α reductase is an important enzyme related with secretion of DHT [26]. In the current study, we found that serum and prostatic DHT concentration was suppressed in HR-treated rats compared with that in the BPH group. Moreover, the mRNA level of 5-α reductase in HR-treated rats was less than that in the rats with BPH. Finasteride, a type II 5α-reductase inhibitor, suppresses both the plasma and intraprostatic DHT concentrations, leading to reductions in glandular size, epithelial cell height and the synthetic and proliferative activities of DHT [27]. Our findings indicate that HR can suppress the DHT level, via inhibition of the action of 5a-reductase, but exact mechanisms should be elucidated through further studies.

We further investigated the mechanism by which HR prevents the development of BPH. The main pathological characteristic of BPH is prostatic epithelial and stromal cell hyperplasia, which is caused by an imbalance between cell proliferation and cell death [28]. In BPH, prostate tissue shows decreased apoptosis due to the loss of control of proliferation and apoptosis balance [29]. PCNA is a nuclear protein that is upregulated during G1/S phase transition and is used to identify proliferating cells [30]. Using PCNA, we confirmed the cell proliferation level in the investigated prostate tissue. Apoptosis is a programmed cell death involved in eliminating cells, and there are numerous pro-apoptotic proteins. Among these, caspase-3 is the main caspase that directly induces cell death. In addition, Bax is a pro-apoptotic protein that stimulates mitochondria to release cytochrome c, which promotes caspase-3, an essential enzyme for programmed cell death [31]. In this study, we found that PCNA was abundant in the prostatic cells of

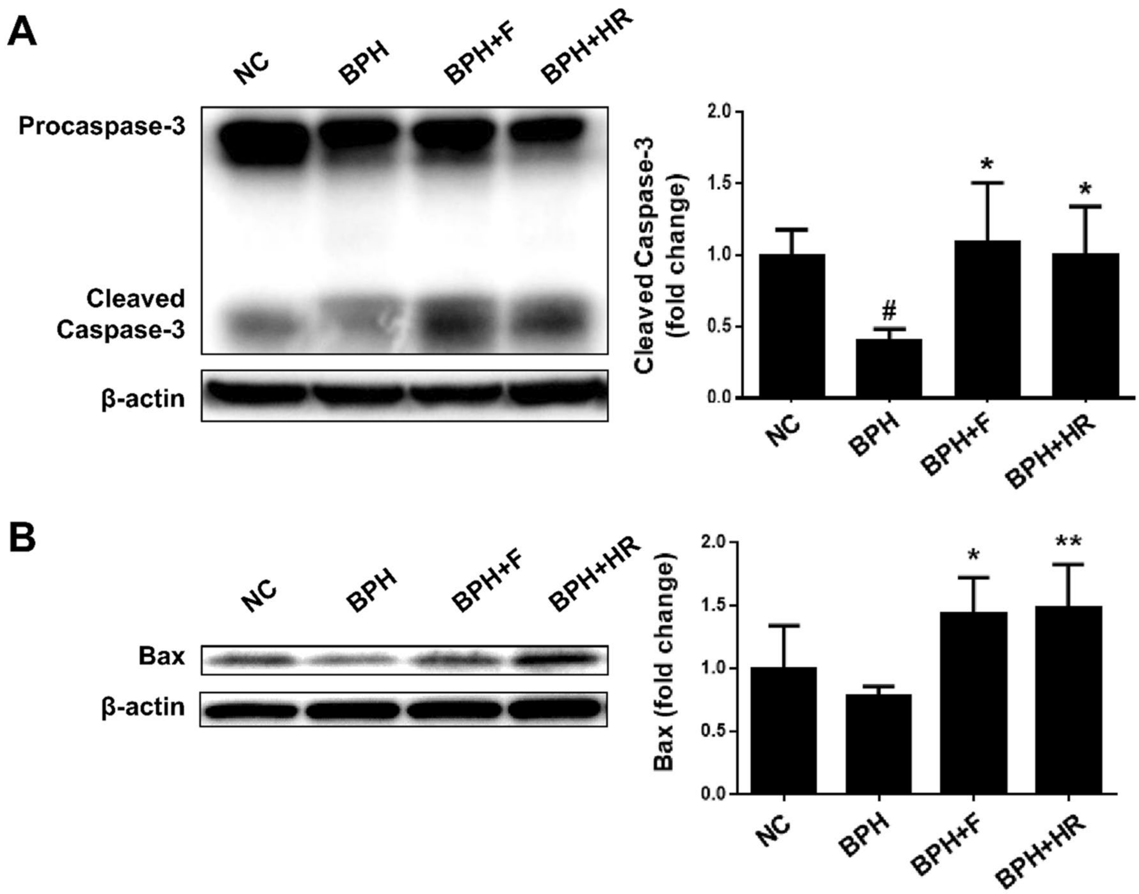

Fig. 5 Effect of HR on the expression of apoptosis-related proteins in prostates of rats with BPH. **A** Expression of caspase 3 and **B** expression of Bax, detected using western blotting. *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P < 0.05$ compared with the NC group; $^{*}P < 0.05$ and $^{**}P < 0.01$ compared with the BPH group

rats with BPH, and the administration of HR reduced this expression in the prostates of these rats. Furthermore, HR-treated animals exhibited higher cleaved caspase-3 and Bax levels in comparison with rats in the BPH group, suggesting that the inhibitory effects of HR on BPH development may comprise both anti-proliferative and pro-apoptotic activities.

Inflammation is considered to have an important role in the pathogenesis of BPH [32]. Accordingly, immune cells infiltrated the prostates in BPH, including T-lymphocytes, macrophages, and B-lymphocytes. These cells are typically chronically active and secrete cytokines such as TNF-α, IL-6, and IL-8, which promote proliferation throughout BPH progression [33, 34]. TNF-α is a versatile cytokine, and it promotes inflammation and immunological response. Under inflammatory condition in BPH, TNF-α may contribute to prostate overgrowth [35]. IL-6 develops the proliferation of prostatic epithelial cells and is secreted by T-lymphocytes, macrophages, and epithelial cells in patients with BPH [36]. IL-8 can stimulate the production of fibroblast growth factor-2, a strong growth factor of stromal and epithelial cells, and consequently enhance the aberrant proliferation of prostatic cells. It is secreted by epithelial prostate cells [10]. Here, the TNF-α, IL-6, and IL-8 levels were higher in the prostates of the BPH group than in the prostates of the NC group. In addition, HR significantly reduced these cytokine expression levels, revealing that HR may decrease BPH by modulating the inflammatory response.

Prostatic inflammation triggers oxidative stress, which leads to the NF-κB pathway, which adjusts inflammation, proliferation, and apoptosis. NF-κB translocates to the nucleus, where it induces the transcription of iNOS and COX-2, all of which are crucial factors in defining the relationship between inflammation and prostate growth [37]. iNOS is mainly responsible for generating reactive nitrogen oxide, which damages prostate glandular epithelium [38]. COX-2 is expressed in all inflammatory cells of the prostatic epithelium and interstitium in BPH [39], and it stimulates Bcl-2, which enhances cell proliferation and decreases apoptosis [40]. In addition, COX-2 inhibition can significantly promote prostate cell apoptosis [41].

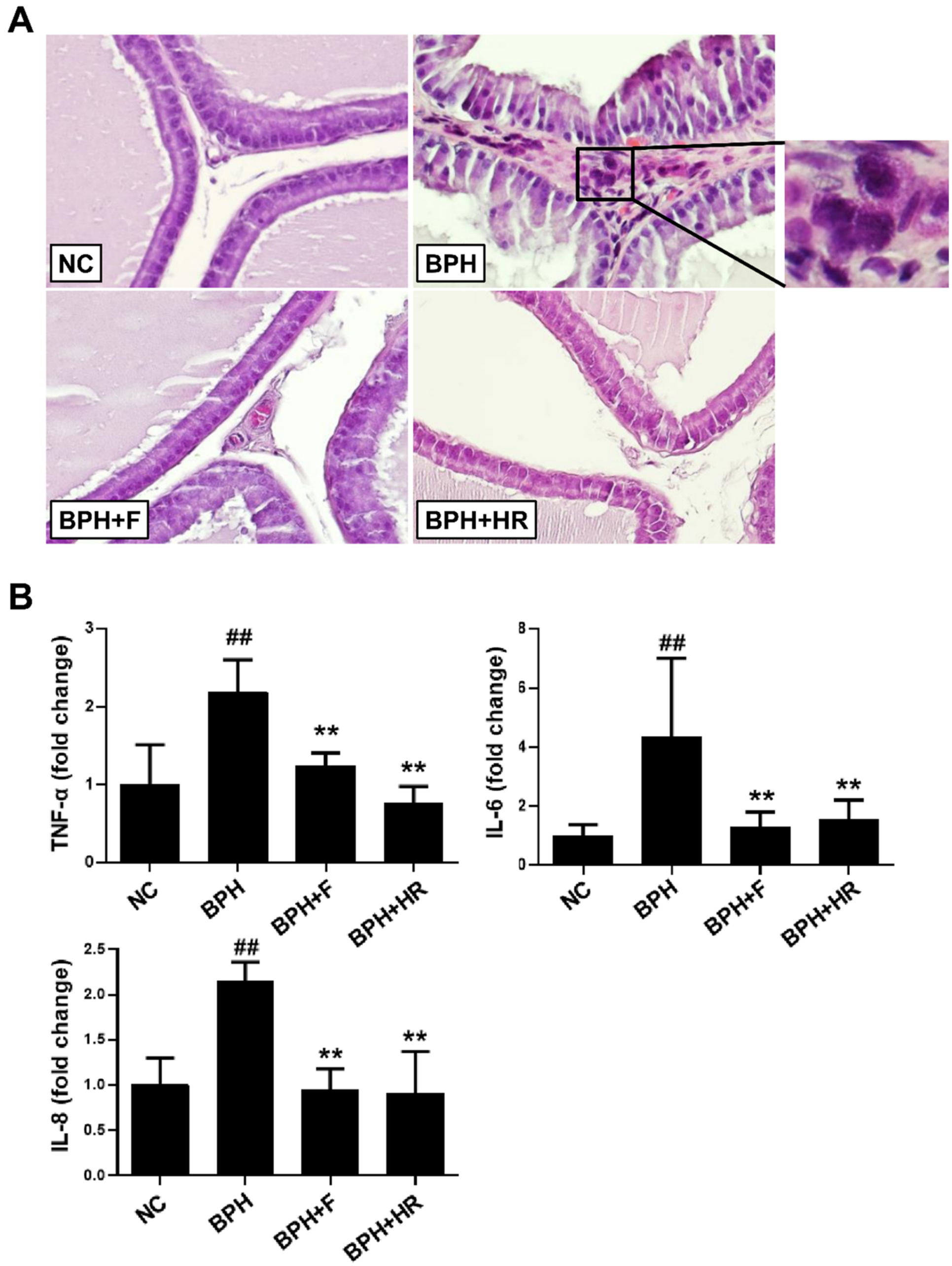

Fig. 6 Effect of HR on inflammation in prostates of rats with BPH. **A** H&E-stained prostate tissue (×400). Inflammatory cell infiltration in the BPH. **B** Inflammatory cytokine levels determined using qRT-PCR. *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ compared with the NC group; $^{*}P < 0.05$ and $^{**}P < 0.01$ compared with the BPH group

Furthermore, NF-κB may induce inflammasome that leads to pyroptosis. A pro-inflammatory form of cell lysis [42]. Pyroptosis in turn releases the mature IL-1β and IL-18 as well as intracellular DAMPs that propagate inflammation and further induces compensative growth of both epithelial and stromal cells and enlargement of prostate [43]. Our findings indicated that NF-κB phosphorylation was elevated in the prostate tissue of rats from the BPH group, which was associated with high levels of iNOS and COX-2, compared with that in the NC group. In comparison to rats in the BPH group, HR-treated rats had a significant lower level of phosphorylated NF-κB, as well as iNOS and COX-2.

Conclusions

This study demonstrated that HR treatment of rats inhibited the development of testosterone-induced BPH by minimizing prostate weight and hyperplasia. This effect can be attributed, at least in part, to an HR-mediated reduction

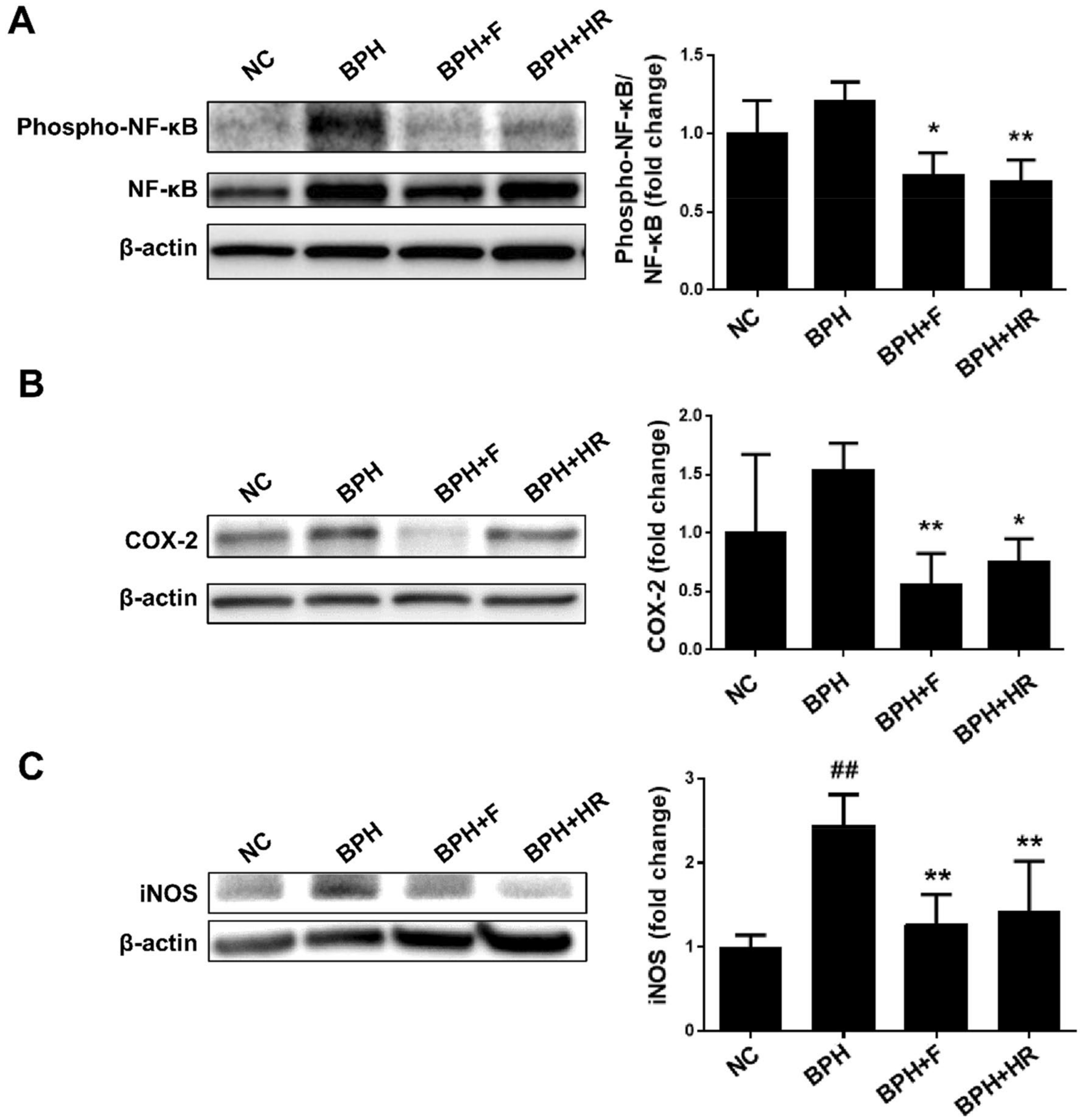

Fig. 7 Effect of HR on the NF-κB pathway in prostates of rats with BPH. Western blotting of NF-κB pathway protein expression. **A** Cytosolic and nuclear protein fractions of prostate for NF-κB expression, **B** COX-2 expression, and **C** iNOS expression. *NC* corn oil injection with PBS; BPH, TP (3 mg/kg) with PBS; BPH + F, TP (3 mg/kg) and finasteride (10 mg/kg); BPH + HR, TP (3 mg/kg) and HR (150 mg/kg). Data are expressed as mean ± SD. $^{\#}P<0.05$ and $^{\#\#}P<0.01$ compared with the NC group; $^{*}P<0.05$ and $^{**}P<0.01$ compared with the BPH

in DHT levels and 5-α reductase in the prostate as well as anti-proliferative, pro-apoptotic, and anti-inflammatory properties of HR. This research suggests that HR has the potential to be used as new therapeutic agents for treatment of BPH; nonetheless, due to the limitation such as small sample size, variability in sample harvest, not having knock-out, and no voiding/functional data, more research is needed to evaluate the clinical value of HR.

Author contributions Conceptualization: SP, Y-SW, and H-JK. Preparation and analysis of extracts: Y-HH. Data curation: SP, E-BB, and E-JH. Investigation: SP, Y-HH, Y-SW, and H-JK. Writing—original draft preparation: SP, Y-HH, and Y-SW. Writing—review and editing: SP, Y-SW, and H-JK. Supervision: Y-SW and H-JK. All authors have read and agreed to the published version of the manuscript.

Funding This work was supported by research fund of Chungnam National University.

Data availability statement Not applicable.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

Ethical approval All protocols for the studies were approved by the Animal Experimental Ethics Committee of Chungnam National University (Daejeon, South Korea; Approval No. CNU-00711).

References

1. Minutoli L et al (2016) Apoptotic pathways linked to endocrine system as potential therapeutic targets for benign prostatic hyperplasia. Int J Mol Sci 17:1311. https://doi.org/10.3390/ijms17081311
2. Abdelwahab O et al (2012) Evaluation of the resistive index of prostatic blood flow in benign prostatic hyperplasia. Int Braz J Urol 38:250–257. https://doi.org/10.1590/S1677-55382012000200014
3. Patel ND, Parsons JK (2014) Epidemiology and etiology of benign prostatic hyperplasia and bladder outlet obstruction. Indian J Urol 30:170–176. https://doi.org/10.4103/0970-1591.126900
4. Ub Wijerathne C et al (2017) *Quisqualis indica* improves benign prostatic hyperplasia by regulating prostate cell proliferation and apoptosis. Biol Pharm Bull 40:2125–2133. https://doi.org/10.1248/bpb.b17-00468
5. Izumi K et al (2013) Androgen receptor roles in the development of benign prostate hyperplasia. Am J Pathol 182:1942–1949. https://doi.org/10.1016/j.ajpath.2013.02.028
6. Andriole G et al (2004) Dihydrotestosterone and the prostate: the scientific rationale for 5alpha-reductase inhibitors in the treatment of benign prostatic hyperplasia. J Urol 172:1399–1403. https://doi.org/10.1097/01.ju.0000139539.94828.29
7. Ishimaru T, Pages L, Horton R (1977) Altered metabolism of androgens in elderly men with benign prostatic hyperplasia. J Clin Endocrinol Metab 45:695–701. https://doi.org/10.1210/jcem-45-4-695
8. Feldman HA et al (2002) Age trends in the level of serum testosterone and other hormones in middle-aged men: longitudinal results from the Massachusetts male aging study. J Clin Endocrinol Metab 87:589–598. https://doi.org/10.1210/jcem.87.2.8201
9. Liao CH et al (2012) Significant association between serum dihydrotestosterone level and prostate volume among Taiwanese men aged 40–79 years. Aging Male 15:28–33. https://doi.org/10.3109/13685538.2010.550660
10. Gandaglia G et al (2013) The role of chronic prostatic inflammation in the pathogenesis and progression of benign prostatic hyperplasia (BPH). BJU Int 112:432–441. https://doi.org/10.1111/bju.12118
11. Bostanci Y et al (2013) Correlation between benign prostatic hyperplasia and inflammation. Curr Opin Urol 23:5–10. https://doi.org/10.1097/MOU.0b013e32835abd4a
12. Ren H et al (2015) The effects of ROS in prostatic stromal cells under hypoxic environment. Aging Male 18:84–88. https://doi.org/10.3109/13685538.2015.1018159
13. Verze P, Cai T, Lorenzetti S (2016) The role of the prostate in male fertility, health and disease. Nat Rev Urol 13:379–386. https://doi.org/10.1038/nrurol.2016.89
14. Huang SS et al (2008) Antioxidant and antiproliferative activities of the four Hydrocotyle species from Taiwan. Bot Stud 49:311–322
15. Kwon HC, Zee OP, Lee KR (1998) Two new monogalactosylacylglycerols from *Hydrocotyle ramiflora*. Planta Med 64:477–479. https://doi.org/10.1055/s-2006-957491
16. Yang JS et al (2007) Effect of alpha-spinasterol extracted from *Phytolacca americanna* on the apoptosis of U937 cell line. J Physiol Pathol Korean Med 21:1108–1117
17. Jeong SI et al (2004) Alpha-spinasterol isolated from the root of *Phytolacca americana* and its pharmacological property on diabetic nephropathy. Planta Med 70:736–739. https://doi.org/10.1055/s-2004-827204
18. Borges FR et al (2014) Anti-inflammatory action of hydroalcoholic extract, dichloromethane fraction and steroid α-spinasterol from *Polygala sabulosa* in LPS-induced peritonitis in mice. J Ethnopharmacol 151:144–150. https://doi.org/10.1016/j.jep.2013.10.009
19. Bailey JA, Vincent GG, Burden RS (1976) The antifungal activity of glutinosone and capsidiol and their accumulation in virus-infected tobacco species. Physiol Plant Pathol 8:35–41. https://doi.org/10.1016/0048-4059(76)90005-9
20. Yang EJ, Song KS (2021) The ameliorative effects of capsidiol isolated from elicited *Capsicum annuum* on mouse splenocyte immune responses and neuroinflammation. Phytother Res 35:1597–1608. https://doi.org/10.1002/ptr.6927
21. Kim YJ et al (2021) Inhibitory effects of Gyeji-tang on MMP-9 activity and the expression of adhesion molecules in IL-4- and TNF-α-stimulated BEAS-2B cells. Plants (Basel) 10:951. https://doi.org/10.3390/plants10050951
22. Kumari S et al (2016) Rapid screening and identification of phenolic antioxidants in *Hydrocotyle sibthorpioides* Lam. by UPLC–ESI-MS/MS. Food Chem 203:521–529. https://doi.org/10.1016/j.foodchem.2016.02.101
23. Hoke GP, McWilliams GW (2008) Epidemiology of benign prostatic hyperplasia and comorbidities in racial and ethnic minority populations. Am J Med 121:S3–S10. https://doi.org/10.1016/j.amjmed.2008.05.021
24. Wilson EM, French FS (1976) Binding properties of androgen receptors. Evidence for identical receptors in rat testis, epididymis, and prostate. J Biol Chem 251:5620–5629. https://doi.org/10.1016/S0021-9258(17)33103-4
25. Mcconnell JD (1991) The pathophysiology of benign prostatic hyperplasia. J Androl 12:356–363. https://doi.org/10.1002/j.1939-4640.1991.tb00272.x
26. Roehrborn CG (2008) Pathology of benign prostatic hyperplasia. Int J Impot Res 20:S11–S18. https://doi.org/10.1038/ijir.2008.55
27. Azzouni F et al (2012) The 5 alpha-reductase isozyme family: a review of basic biology and their role in human diseases. Adv Urol 2012:530121. https://doi.org/10.1155/2012/530121
28. La Vignera S et al (2016) Endocrine control of benign prostatic hyperplasia. Andrology 4:404–411. https://doi.org/10.1111/andr.12186
29. Kyprianou N, Tu H, Jacobs SC (1996) Apoptotic versus proliferative activities in human benign prostatic hyperplasia. Hum Pathol 27:668–675. https://doi.org/10.1016/S0046-8177(96)90396-2
30. Schönenberger F et al (2015) Discrimination of cell cycle phases in PCNA-immunolabeled cells. BMC Bioinformatics 16:180. https://doi.org/10.1186/s12859-015-0618-9
31. Barman J et al (2018) Apoptosis: mediator molecules, interplay with other cell death processes and therapeutic potentials. Curr Pharm Biotechnol 19:644–663. https://doi.org/10.2174/1389201019666180821093239

32. Ribal MJ (2013) The link between benign prostatic hyperplasia and inflammation. Eur Urol Suppl 12:103–109. https://doi.org/10.1016/j.eursup.2013.08.001
33. Wang L et al (2008) Chronic inflammation in benign prostatic hyperplasia: implications for therapy. Med Hypotheses 70:1021–1023. https://doi.org/10.1016/j.mehy.2007.08.022
34. McLaren ID, Jerde TJ, Bushman W (2011) Role of interleukins, IGF and stem cells in BPH. Differentiation 82:237–243. https://doi.org/10.1016/j.diff.2011.06.001
35. Yang X et al (2014) Abacopteris penangiana exerts testosterone-induced benign prostatic hyperplasia protective effect through regulating inflammatory responses, reducing oxidative stress and anti-proliferative. J Ethnopharmacol 157:105–113. https://doi.org/10.1016/j.jep.2014.09.025
36. De Nunzio C, Presicce F, Tubaro A (2016) Inflammatory mediators in the development and progression of benign prostatic hyperplasia. Nat Rev Urol 13:613–626. https://doi.org/10.1038/nrurol.2016.168
37. Chughtai B et al (2011) Role of inflammation in benign prostatic hyperplasia. Rev Urol 13:147–150
38. Baltaci S et al (2001) Inducible nitric oxide synthase expression in benign prostatic hyperplasia, low-and high-grade prostatic intraepithelial neoplasia and prostatic carcinoma. BJU Int 88:100–103. https://doi.org/10.1046/j.1464-410x.2001.02231.x
39. Wang W, Bergh A, Damber JE (2004) Chronic inflammation in benign prostate hyperplasia is associated with focal upregulation of cyclooxygenase-2, Bcl-2, and cell proliferation in the glandular epithelium. Prostate 61:60–72. https://doi.org/10.1002/pros.20061
40. Nakanishi Y et al (2001) Inhibitors of cyclooxygenase-2 (COX-2) suppressed the proliferation and differentiation of human leukaemia cell lines. Eur J Cancer 37:1570–1578. https://doi.org/10.1016/S0959-8049(01)00160-5
41. Altavilla D et al (2012) Effects of flavocoxid, a dual inhibitor of COX and 5-lipoxygenase enzymes, on benign prostatic hyperplasia. Br J Pharmacol 167:95–108. https://doi.org/10.1111/j.1476-5381.2012.01969.x
42. Inouye BM et al (2018) The emerging role of inflammasomes as central mediators in inflammatory bladder pathology. Curr Urol 11:57–72. https://doi.org/10.1159/000447196
43. Jiang MY et al (2017) Mitochondrion-associated protein peroxiredoxin 3 promotes benign prostatic hyperplasia through autophagy suppression and pyroptosis activation. Oncotarget 8:80295–80302. https://doi.org/10.18632/oncotarget.17927